



BAULKHAM HILLS HIGH SCHOOL

YEAR 12 TASK 2 2023

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 1 hour
- Write using black pen
- Calculators approved by NESA may be used
- Show all necessary working in questions 6-8
- Marks may be deducted for careless or badly arranged work
- Reference sheet is provided

Total marks – 46

Exam consists of 6 pages.

This paper consists of TWO sections.

Section I – (5 marks) Pages (2-3)

Questions 1-5

- Attempt Questions 1-5
- Answer on answer sheet provided
- Allow about 8 minutes for this section

Section II – (41 marks) Pages (4-6)

- Attempt Questions 6-8
- Allow about 52 minutes
- Answer each question on the appropriate page of your answer booklet

Section I - 5 marks

Use the multiple-choice answer sheet for questions 1-5.

1. Which of the following is the derivative of $\log_e(\sin^{-1} x)$?

A) $\frac{1}{\sqrt{1-x^2}}$

B) $\frac{1}{\sqrt{1-x^2} \sin^{-1} x}$

C) $\ln(\sqrt{1-x^2})$

D) $\frac{1}{\sqrt{1-x^2} \cos^{-1} x}$

2. Which of the following is an expression for $\int \cos^2 2x \, dx$?

A) $x - \frac{1}{4} \sin 4x + c$

B) $x + \frac{1}{4} \sin 4x + c$

C) $\frac{x}{2} - \frac{1}{8} \sin 4x + c$

D) $\frac{x}{2} + \frac{1}{8} \sin 4x + c$

3. Which of the following is an expression for $\int \frac{e^x}{1+e^{2x}} \, dx$?

Use the substitution $u = e^x$.

A) $e^x \tan^{-1} e^x + c$

B) $e^x \tan^{-1} e^{2x} + c$

C) $\tan^{-1} e^x + c$

D) $\tan^{-1} e^{2x} + c$

4. Which of the following is the exact volume of the solid formed by rotating the curve $y = \frac{4}{x-2}$ about the x -axis, between $x = 3$ and $x = 6$?

- A) $4\pi \ln 4$
- B) $16\pi \ln 4$
- C) 6π
- D) 12π

5. What is the exact value of $\int_{\frac{\pi}{2}}^{\pi} (\sin^2 x + x) dx$?

- A) $\frac{3\pi^2 + \pi + 2}{8}$
- B) $\frac{3\pi^2 + \pi}{8}$
- C) $\frac{3\pi^2 + 2\pi + 2}{8}$
- D) $\frac{3\pi^2 + 2\pi}{8}$

End of Section I

Section II

41 marks

Attempt Questions 6-8

Allow about 52 minutes for this section

Answer each question on the appropriate page of your answer booklet. Extra paper is available.

Your responses should include relevant calculations and/or mathematical reasoning.

Question 6 (14 marks)

Marks

a) i. Show that $\sin\left(x + \frac{\pi}{4}\right) + \sin\left(x - \frac{\pi}{4}\right) = \sqrt{2} \sin x$. 2

ii. Hence, or otherwise, solve $\sin\left(x + \frac{\pi}{4}\right) + \sin\left(x - \frac{\pi}{4}\right) = 1$ for $0 \leq x \leq \pi$. 1

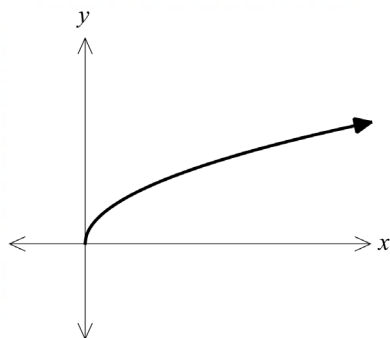
b) Use the substitution $u = \tan x$ to find $\int_0^{\frac{\pi}{4}} \tan^2 x \sec^2 x \, dx$. 3

c) Find $\int_0^1 \frac{dx}{\sqrt{9-4x^2}}$, leaving your answer in exact form. 2

d) Differentiate $e^{3x} \tan^{-1}\left(\frac{x}{2}\right)$. 2

e) Evaluate $\int_0^{\frac{\pi}{8}} \sin^2 3x \, dx$. 3

f) Sketch the solid formed when the region between the curve $y = \sqrt{x}$ and the y -axis from $y = 0$ to $y = 4$ is rotated about the y -axis. 1



Question 7 (13 marks)

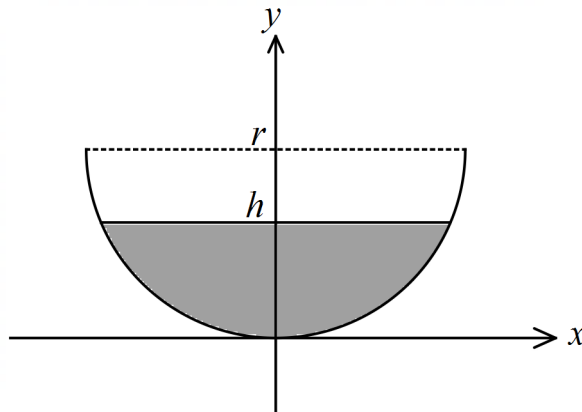
- a) i. Express $7 \cos \theta - \sin \theta$ in the form $R \cos(\theta + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$. **2**
- ii. Hence solve $7 \cos \theta - \sin \theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$, giving answers to the nearest degree. **2**

b) If $\int_{-2}^{2\sqrt{3}} \frac{dx}{\sqrt{16-x^2}} = k\pi$, find the value of k . **3**

- c) A hemi-spherical bowl of radius r contains water up to the height h , where $0 < h < r$. By considering the hemi-spherical bowl as a solid generated when the semi-circle

$y = r - \sqrt{r^2 - x^2}$ is rotated about the y -axis, prove that the volume of the water in the bowl

is $\frac{\pi h^2(3r-h)}{3}$.



- d) Use the substitution $t = \tan x$ to solve the equation $\sin 2x - 3 \cos 2x = 3$ for $0 \leq x \leq 2\pi$. **3**
- Give your answer correct to two decimal places.

Question 8 (14 marks)**Marks**

a) i. Prove that $\tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$.

3

ii. Hence, or otherwise, find $\int \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x} dx$.

2

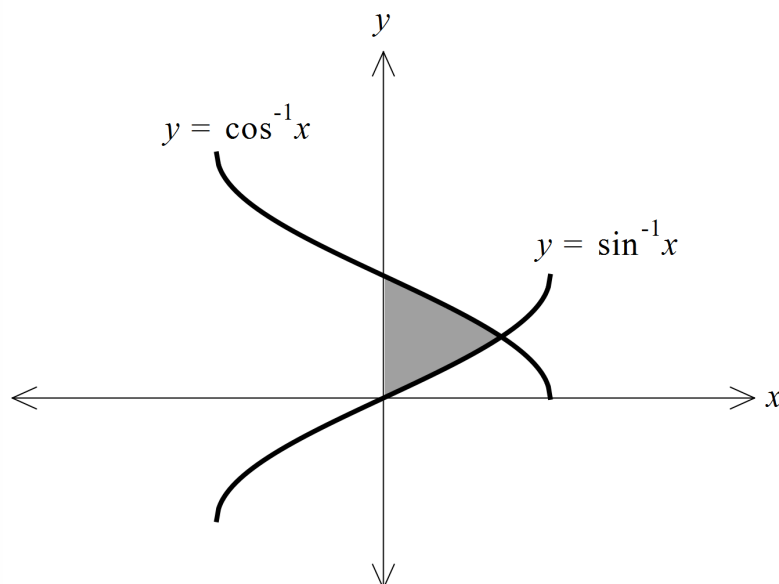
b) By using $u = x - 1$, find $\int \frac{x+1}{\sqrt[3]{x-1}} dx$.

3

c) Use polynomial division, or otherwise, to find the exact value of $\int_0^2 \frac{x^4 + x^2 - 4}{x^2 + 4} dx$.

3

d) Find the exact area bounded by $y = \sin^{-1} x$, $y = \cos^{-1} x$ and the y -axis.

3**End of Exam**

Baulkham Hills High School
Task 2 Examination 2023

Marking Guideline- Yr 12 Mathematics Extension 1

Section I (5 marks)

Award 1 marks to each correct answer.

Answers: 1. B 2. D 3. C 4. D 5. D

Question	Suggested solutions	Answer
1	$\frac{d}{dx} \left(\log_e \left(\sin^{-1} x \right) \right) = \frac{1}{\sqrt{1-x^2} \times \sin^{-1} x}$	B
2	$\int \cos^2 2x \, dx$ $= \int \frac{1}{2} (1 + \cos 4x) \, dx$ $= \frac{1}{2} \left(x + \frac{1}{4} \sin 4x \right) + c$ $= \frac{x}{2} + \frac{1}{8} \sin 4x + c$	D
3	$\int \frac{e^x}{1+e^{2x}} dx$ $= \int \frac{du}{1+u^2} \quad \begin{matrix} u = e^x \\ dx = \frac{du}{e^x} \end{matrix}$ $= \tan^{-1} u + c$ $= \tan^{-1} e^x + c$	C
4	$V = \pi \int_3^6 \left(\frac{4}{x-2} \right)^2 dx$ $= 16\pi \int_3^6 (x-2)^{-2} dx$ $= -16\pi \left[(x-2)^{-1} \right]_3^6$ $= -16\pi \left[(6-2)^{-1} - (3-2)^{-1} \right]$ $= 12\pi$	D

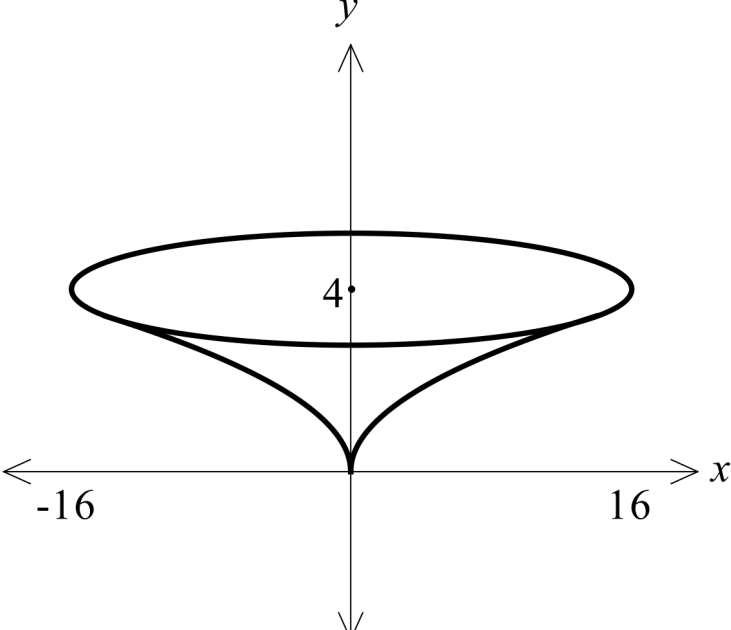
5	$\int_{\frac{\pi}{2}}^{\pi} (\sin^2 x + x^2) \, dx$ $= \int_{\frac{\pi}{2}}^{\pi} \left[\frac{1}{2}(1 - \cos 2x) + x \right] dx$ $= \left[\frac{1}{2}x - \frac{1}{4}\sin 2x + \frac{x^2}{2} \right]_{\frac{\pi}{2}}^{\pi}$ $= \left(\frac{1}{2}\pi - \frac{1}{4}\sin 2\pi + \frac{\pi^2}{2} \right) - \left(\frac{1}{2} \times \frac{\pi}{2} - \frac{1}{4}\sin \pi + \frac{\pi^2}{8} \right)$ $= \frac{3\pi^2 + 2\pi}{8}$	D
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Section II (40 marks)

In all questions, award full marks for correct answers with necessary working.

Use the suggested solutions in conjunction with the marking criteria.

Q	Suggested Solutions	Marking criteria
6ai	$\sin\left(x + \frac{\pi}{4}\right) + \sin\left(x - \frac{\pi}{4}\right) = \sin x \cos \frac{\pi}{4} + \cos x \sin \frac{\pi}{4} + \sin x \cos \frac{\pi}{4} - \cos x \sin \frac{\pi}{4}$ $= \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x - \frac{1}{\sqrt{2}} \cos x$ $= \frac{2}{\sqrt{2}} \sin x$ $= \sqrt{2} \sin x$	2- correct solution 1- expands using the compound angle formula or equivalent
6aii	$\sin\left(x + \frac{\pi}{4}\right) + \sin\left(x - \frac{\pi}{4}\right) = 1$ $\sqrt{2} \sin x = 1$ $\sin x = \frac{1}{\sqrt{2}}$ $x = \frac{\pi}{4}, \frac{3\pi}{4}$	1- correct solution
6b	$\int_0^{\frac{\pi}{4}} \tan^2 x \sec^2 x \, dx \quad u = \tan x$ $\frac{du}{dx} = \sec^2 x$ $du = \sec^2 x \, dx$ $= \int_0^1 u^2 \, du$ $= \left[\frac{u^3}{3} \right]_0^1$ $= \frac{1}{3}$ $u = \tan \frac{\pi}{4} \quad u = \tan 0$ $u = 1 \quad u = 0$	3- correct solution 2- uses the substitution method correctly and correctly identifying the limits 1- correctly identifies the new limits or uses the substitution method correctly
6c	$\int_0^1 \frac{dx}{\sqrt{9-4x^2}}$ $= \int_0^1 \frac{dx}{\sqrt{3^2 - 2^2 x^2}}$ $= \frac{1}{2} \left[\sin^{-1} \left(\frac{2x}{3} \right) \right]_0^1$ $= \frac{1}{2} \sin^{-1} \frac{2}{3}$	2- correct solution 1- obtains the inverse $R \sin^{-1} \left(\frac{2x}{3} \right)$

6d	$\frac{d}{dx}\left(e^{3x} \tan^{-1}\left(\frac{x}{2}\right)\right)$ $v = \tan^{-1}\left(\frac{x}{2}\right)$ $u = e^{3x}$ $u' = 3e^{3x} \quad v' = \frac{1}{2} \times \frac{1}{1 + \left(\frac{x}{2}\right)^2}$ $v' = \frac{2}{x^2 + 4}$ $u'v + v'u = 3e^{3x} \tan^{-1}\left(\frac{x}{2}\right) + \frac{2e^{3x}}{x^2 + 4}$	2- correct solution 1- correct differentiates $\tan^{-1} \frac{x}{2}$
6e	$\int_0^{\frac{\pi}{8}} \sin^2 3x \, dx$ $= \int_0^{\frac{\pi}{8}} \left(\frac{1}{2} - \frac{1}{2} \cos(2 \times 3x) \right) dx$ $= \int_0^{\frac{\pi}{8}} \left(\frac{1}{2} - \frac{1}{2} \cos 6x \right) dx$ $= \left[\frac{x}{2} - \frac{1}{12} \sin 6x \right]_0^{\frac{\pi}{8}}$ $= \left(\frac{\pi}{16} - \frac{1}{12} \sin \left(\frac{\pi}{8} \times 6 \right) \right) - \left(0 - \frac{1}{12} \sin 0 \right)$ $= \frac{\pi}{16} - \frac{1}{12\sqrt{2}}$	3- correct solution 2- obtains the correct primitive 1- transforms the integrand
6f		1- correct graph

7ai	$R \cos(\theta + \alpha) = R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$ $7 \cos \theta - \sin \theta \equiv R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$ $R \cos \alpha = 7 \quad R \sin \alpha = 1$ $\cos \alpha = \frac{7}{R} \quad \sin \alpha = \frac{1}{R}$ $R = \sqrt{1^2 + 7^2}$ $R = \sqrt{50}$ $\tan \alpha = \frac{1}{7}$ $\alpha = 8^\circ 7' 48.37''$ $\therefore 7 \cos \theta - \sin \theta = 5\sqrt{2} \cos(\theta + 8^\circ)$	2- correct solution 1- correctly finds R or α
7a ii	$5\sqrt{2} \cos(\theta + 8^\circ) = 5$ $\cos(\theta + 8^\circ) = \frac{1}{\sqrt{2}}$ $\theta + 8^\circ = 45^\circ, 315^\circ \quad 0^\circ \leq \theta \leq 360^\circ$ $\therefore \theta = 37^\circ, 307^\circ \quad 8^\circ \leq \theta + 8^\circ \leq 368^\circ$	2- correct solution 1- correctly finds 1 angle/solution
7b	$\int_{-2}^{2\sqrt{3}} \frac{dx}{\sqrt{16-x^2}} = k\pi$ $\left[\sin^{-1}\left(\frac{x}{4}\right) \right]_{-2}^{2\sqrt{3}}$ $= \sin^{-1}\left(\frac{2\sqrt{3}}{4}\right) - \sin^{-1}\left(\frac{-2}{4}\right)$ $= \frac{\pi}{2}$ $\therefore k = \frac{1}{2}$	3- correct solution 2- correctly integrates for sin inverse and substitutes the values in 1- correctly integrates for sin inverse
7c	$y = r - \sqrt{r^2 - x^2}$ $(y - r)^2 = r^2 - x^2$ $\therefore x^2 = r^2 - (y - r)^2$ $x^2 = 2ry - y^2$ $V = \pi \int_0^h (2ry - y^2) dy$ $= \pi \left[ry^2 - \frac{y^3}{3} \right]_0^h$ $= \pi \left(rh^2 - \frac{h^3}{3} \right)$ $= \pi h^2 \left(r - \frac{h}{3} \right)$ $= \frac{\pi h^2 (3r - h)}{3} \text{ units}^3$	3- correctly obtains the answer 2- obtains the correct integrand to find the volume 1- finds the correct equation for x^2

7d	let $t = \tan x$ $\left(\frac{2t}{1+t^2}\right) - 3\left(\frac{1-t^2}{1+t^2}\right) = 3$ $2t - 3 + 3t^2 = 3 + 3t^2$ $t = 3$ $\tan x = 3$ $x = 1.25, 4.39$ test for $\frac{\pi}{2}$ and $\frac{3\pi}{2}$ $\therefore x = 1.25, 4.39, \frac{\pi}{2}, \frac{3\pi}{2}$	3- correct solution 2- finds 2 out of the 4 solutions 1- solves for t
8ai	$\tan 3x = \tan(2x + x)$ $= \frac{\tan 2x + \tan x}{1 - \tan 2x \tan x}$ $= \left(\frac{2 \tan x}{1 - \tan^2 x} + \tan x\right) \div \left(1 - \frac{2 \tan^2 x}{1 - \tan^2 x}\right)$ $= \frac{2 \tan x + \tan x - \tan^3 x}{1 - \tan^2 x - 2 \tan^2 x}$ $= \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$	3- uses correct methods 2- uses compound angle formula and double angle formula correctly 1- correctly uses the compound angle formula or equivalent
8aii	$\int \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x} dx = \int \tan 3x dx$ $\int \tan 3x dx = \int \frac{\sin 3x}{\cos 3x}$ $= -\frac{1}{3} \ln \cos 3x + c$	2- correct solution 1- correctly transforms the integrand to $\frac{\sin 3x}{\cos 3x}$
8b	$\int \frac{x+1}{\sqrt[3]{x-1}} dx$ $= \int \frac{2+u}{\sqrt[3]{u}} du$ $= \int \left(2u^{-\frac{1}{3}} + u^{\frac{2}{3}}\right) du$ $= 3u^{\frac{2}{3}} + \frac{3u^{\frac{5}{3}}}{5} + c$ $= 3(x-1)^{\frac{2}{3}} + \frac{3(x-1)^{\frac{5}{3}}}{5} + c$ $u = x - 1 \quad x = u + 1$ $\frac{du}{dx} = 1$ $du = dx$	3- correct solution 2- correct primitive 1- obtains the correct substitution for u and du

8c	Using polynomial division: $\frac{x^4 + x^2 - 4}{x^2 + 4} = x^2 - 3 + \frac{8}{x^2 + 4}$ $\int_0^2 x^2 - 3 + \frac{8}{x^2 + 4} dx$ $= \left[\frac{x^3}{3} - 3x + \frac{8}{2} \tan^{-1} \frac{x}{2} \right]_0^2$ $= \frac{2^3}{3} - 3(2) + 4 \tan^{-1} \left(\frac{2}{2} \right)$ $= \pi - \frac{10}{3}$	3- correct solution 2- correct primitive 1- transforms integrand into partial fraction
8d	$y = \cos^{-1} x \quad y = \sin^{-1} x$ $x = \cos y \quad x = \sin y$ $\cos y = \sin y$ $\tan y = 1$ $y = \frac{\pi}{4}$ $\int_0^{\frac{\pi}{4}} \sin y \, dy + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos y \, dy$ $= [-\cos y]_0^{\frac{\pi}{4}} + [\sin y]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$ $= -\cos \frac{\pi}{4} + \cos 0 + \sin \frac{\pi}{2} - \sin \frac{\pi}{4}$ $= -\frac{\sqrt{2}}{2} + 1 + 1 - \frac{\sqrt{2}}{2}$ $= 2 - \sqrt{2}$	3- correct solution 2-forms the correct integrand 1-finds $y = \frac{\pi}{4}$ or $x = \frac{1}{\sqrt{2}}$